

HARBIN TAIPING, China

WMO: 541611

Lat: 45.623N

Lon: 126.250E

Elev: 139

StdP: 99.66

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-27.9	-33.9	0.2	-30.1	-31.8	0.2	-27.5	8.4	-9.6	7.5	-10.6	1.7	200	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.9	21.1	30.1	21.6	28.9	21.5	25.1	28.0	24.2	27.5	23.3	26.7	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.3	27.3	23.1	18.1	26.2	22.1	17.1	25.2	77.1	27.4	73.4	27.6	69.7	26.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.3	8.9	7.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-33.4	35.1	4.3	2.0	-36.5	36.5	-39.0	37.7	-41.5	38.8	-44.6	40.2		
			-33.6	26.3	4.3	1.3	-36.6	27.3	-39.1	28.0	-41.5	28.7	-44.6	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-18.4	-13.7	-3.6	6.9	15.0	21.0	23.1	21.6	15.3	6.5	-5.2	-15.8	(6)
(7)		DBStd	15.43	4.94	5.85	6.46	5.11	4.24	3.30	2.23	2.73	3.69	5.00	6.45	5.47	(7)
(8)		HDD10.0	3489	882	665	422	121	6	0	0	5	132	457	801		(8)
(9)		HDD18.3	5435	1140	898	679	344	122	11	1	102	367	706	1059		(9)
(10)		CDD10.0	1474	0	0	1	27	160	329	407	361	165	24	1	0	(10)
(11)		CDD18.3	378	0	0	0	1	18	91	149	108	12	1	0	0	(11)
(12)		CDH23.3	3339	0	0	0	25	272	892	1181	831	132	7	0	0	(12)
(13)		CDH26.7	953	0	0	0	4	87	324	328	197	14	0	0	0	(13)
(14)	Wind	WSAvg	3.5	2.9	3.4	3.9	4.7	4.5	3.5	3.0	2.8	3.1	3.6	3.7	3.2	(14)
(15)	Precipitation	PrecAvg	522	4	5	12	21	51	97	138	103	45	26	13	7	(15)
(16)		PrecMax	816	12	20	32	61	123	183	342	212	81	57	38	19	(16)
(17)		PrecMin	359	0	0	1	1	14	30	62	38	6	9	3	1	(17)
(18)		PrecStd	100	3	5	9	15	29	41	56	43	22	12	10	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.9	5.2	17.0	26.9	32.1	35.9	32.9	32.0	28.2	24.2	14.2	4.0	(19)
(20)		MCWB	-4.0	0.9	8.4	12.3	17.0	20.1	23.9	24.9	19.7	15.1	9.2	1.7		(20)
(21)		2%	DB	-5.1	2.2	12.8	22.2	29.1	32.2	31.0	30.0	26.2	21.1	11.0	-1.8	(21)
(22)		MCWB	-6.6	-0.7	5.2	10.6	15.2	19.7	23.4	23.4	18.5	13.8	7.1	-3.1		(22)
(23)		5%	DB	-7.9	-0.2	9.1	19.9	26.2	30.1	29.8	28.9	24.9	17.9	7.2	-4.2	(23)
(24)		MCWB	-9.1	-2.7	3.3	9.5	14.6	19.9	23.2	22.9	18.1	11.2	4.0	-5.5		(24)
(25)		10%	DB	-10.1	-3.9	6.8	16.9	23.9	28.1	28.8	27.8	23.0	15.1	4.1	-7.1	(25)
(26)		MCWB	-11.1	-5.9	1.7	8.1	13.8	19.0	22.9	22.1	16.3	9.6	1.5	-8.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.2	2.0	8.3	13.4	18.9	23.6	26.2	26.6	21.9	16.9	10.7	1.6	(27)
(28)		MCDWB	-2.7	4.1	15.6	22.5	27.5	28.5	29.7	29.3	25.5	20.4	13.8	3.2		(28)
(29)		2%	WB	-6.8	-0.4	5.6	11.8	17.2	22.2	25.2	25.3	20.0	14.1	6.6	-2.9	(29)
(30)		MCDWB	-5.4	2.2	11.2	20.5	25.9	27.5	28.4	28.0	24.3	20.1	9.6	-1.9		(30)
(31)		5%	WB	-9.1	-3.1	3.6	10.2	15.7	21.2	24.3	24.4	18.8	12.1	4.1	-5.9	(31)
(32)		MCDWB	-8.1	-0.2	8.8	17.8	23.4	26.4	27.6	27.3	23.3	16.4	7.3	-4.5		(32)
(33)		10%	WB	-11.3	-5.7	1.7	8.7	14.6	20.4	23.6	23.2	17.7	10.4	1.6	-8.4	(33)
(34)		MCDWB	-10.3	-3.6	5.8	15.7	22.3	25.7	27.1	26.1	21.5	14.5	3.8	-7.4		(34)
(35)	Mean Daily Temperature Range	MDBR	12.0	13.7	12.6	13.3	12.6	10.9	9.3	10.1	12.7	12.8	10.6	10.6		(35)
(36)		5% DB	MCDBR	13.2	13.9	15.1	17.5	16.5	14.5	11.4	11.5	14.2	15.4	13.1	12.3	(36)
(37)		MCWBR	12.1	11.3	9.4	9.3	7.6	5.8	5.4	5.9	7.5	9.1	9.4	11.2		(37)
(38)		5% WB	MCDBR	13.1	13.4	14.4	14.9	13.8	10.7	9.1	9.4	12.3	14.0	12.6	12.2	(38)
(39)		MCWBR	12.0	11.2	9.2	8.8	6.9	5.2	5.3	5.6	7.3	9.1	9.6	11.2		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.324	0.340	0.416	0.517	0.581	0.627	0.587	0.516	0.462	0.458	0.375	0.335		(40)
(41)		taud	2.083	2.070	1.946	1.781	1.703	1.703	1.824	1.982	2.028	1.924	2.049	2.063		(41)
(42)		Ebn at Noon	739	812	793	744	711	680	703	740	742	661	663	671		(42)
(43)		Edh at Noon	98	121	159	206	231	232	203	167	146	140	102	91		(43)
(44)	All-Sky Solar Radiation	RadAvg	2.10	3.22	4.30	4.96	5.51	5.62	5.16	4.85	4.30	3.00	1.97	1.67		(44)
(45)		RadStd	0.20	0.23	0.32	0.35	0.49	0.54	0.48	0.35	0.34	0.19	0.15	0.10		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page